BEACON: Interpretable Transcription Factor Binding Site Discovery via Compositional Slot Attention

Introduction

Identifying transcription factor (TF) binding sites from ChIP-seq data is central to understanding gene regulation, with direct implications for variant interpretation, drug target discovery, and synthetic biology. Deep learning approaches such as BPNet [1] and ChromBPNet [2] achieve high accuracy for binding profile prediction, but rely on post-hoc attribution methods (DeepSHAP, TF-MoDISco) to extract biological insight from learned representations. This two-stage predict-then-interpret paradigm has three fundamental limitations: (i) attribution methods introduce artifacts from reference choice and approximation error, (ii) discovered motifs require manual curation and lack confidence scores, and (iii) the approach cannot directly enumerate discrete binding events within a regulatory region—it can highlight *where* signal is important, but not decompose a multi-TF locus into individual binding events with identity and strength.

We present **BEACON** (**B**inding **E**vent **A**ttention-based **C**ompositional **O**bject **N**etwork), which reconceptualizes TF binding site discovery as an object-centric decomposition problem. Inspired by slot attention mechanisms from computer vision [3], BEACON treats each binding event as a discrete object characterized by its genomic position, TF identity, and occupancy strength. Rather than predicting an aggregate profile and attributing it post-hoc, BEACON directly decomposes regulatory sequences into constituent binding events through competitive attention—providing inherent interpretability while simultaneously improving predictive accuracy over existing methods.

Methods

BEACON consists of four components. (1) A *dilated CNN backbone* (2^0–2^8 dilation rates, 1029 bp receptive field) encodes one-hot DNA sequences (L=2000 bp), capturing both local motif patterns and long-range cooperativity. (2) A *helical positional encoding* captures the 10.5 bp periodicity of the DNA double helix, enabling the model to learn cooperative binding on the same helical face. (3) K=16 learnable *slot vectors* compete for sequence positions via T=3 iterations of slot attention with per-slot softmax and GRU updates; this independent attention mode allows multiple slots to activate simultaneously for multi-TF sequences. (4) *Per-slot heads* predict binding site position (Gaussian μ, σ), TF identity (softmax), and occupancy (sigmoid). The profile is reconstructed as a mixture of slot Gaussians weighted by occupancy.

Training uses a composite loss: profile reconstruction (KL + MSE), Hungarian-matched [4] site position loss (solving permutation invariance), TF classification cross-entropy, and slot diversity regularization. We use chromosome-based splits (chr1–17/18–19/20–22,X) and evaluate on ENCODE ChIP-seq for 7 TFs in K562 (CTCF, GATA1, TAL1, MYC, MAX, SPI1, CEBPB), spanning zinc finger, GATA, bHLH, ETS, and bZIP families (28,812 train / 1,533 val / 1,197 test sequences).

Results

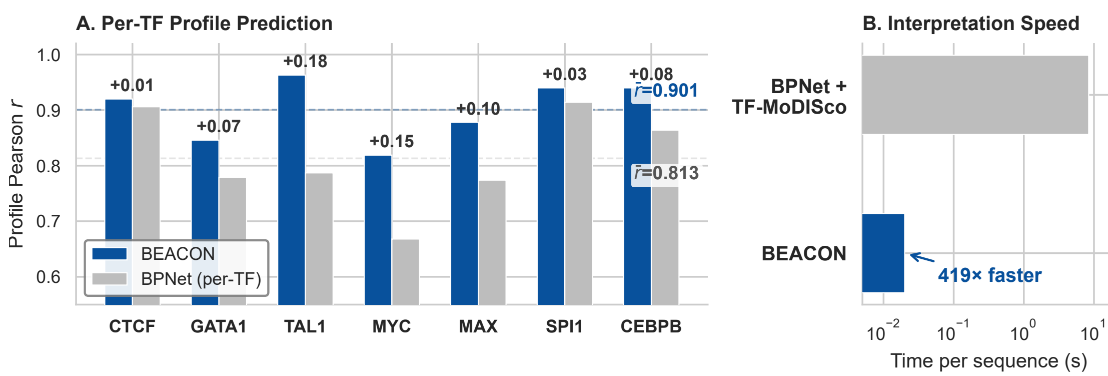

Figure 1: BEACON vs. BPNet on 7 TFs in K562. **(A)** Per-TF profile Pearson r: a single BEACON model (850K params) outperforms 7 separately trained BPNet models (997K total params) on every TF. Dashed lines show mean r across TFs ($\bar{r} = 0.901$ for BEACON vs. $\bar{r} = 0.813$ for BPNet). The largest improvements occur on bHLH factors (TAL1: +0.18, MYC: +0.15), where BEACON's joint multi-TF architecture captures cooperative binding patterns—such as obligate MYC–MAX heterodimerization—that are inaccessible to per-TF models. **(B)** Binding site enumeration speed: BEACON directly outputs structured binding events (position, identity, confidence) in 20 ms per sequence, compared to 8.4 s for BPNet followed by TF-MoDISco post-hoc attribution—a 419× speedup that enables genome-wide binding catalogues in minutes rather than hours.

BEACON achieves mean profile Pearson $r = 0.901$ across 7 TFs, surpassing BPNet ($r = 0.813$) on every TF (Fig. 1A) with a single unified model (850K params) versus BPNet's 7 separately trained per-TF models (997K total). The largest gains are on bHLH factors (TAL1: +0.18, MYC: +0.15), where BEACON's multi-TF architecture captures co-binding patterns—such as obligate MYC–MAX heterodimerization—that single-TF models cannot represent. Binding site enumeration is 419× faster than BPNet+TF-MoDISco (20 ms vs. 8.4 s per sequence; Fig. 1B), enabling genome-wide binding event cataloguing in minutes. Site detection achieves F1 = 0.984 (100% precision, 96.9% recall) and per-TF classification reaches 75.5% overall (5.3× above chance), with

SPI1 and CEBPB exceeding 90% and CTCF at 87%. Within-family confusion (e.g., MYC–MAX, both bHLH) is 3.2× higher than between-family confusion, reflecting genuine biological similarity in binding profiles rather than model failure.

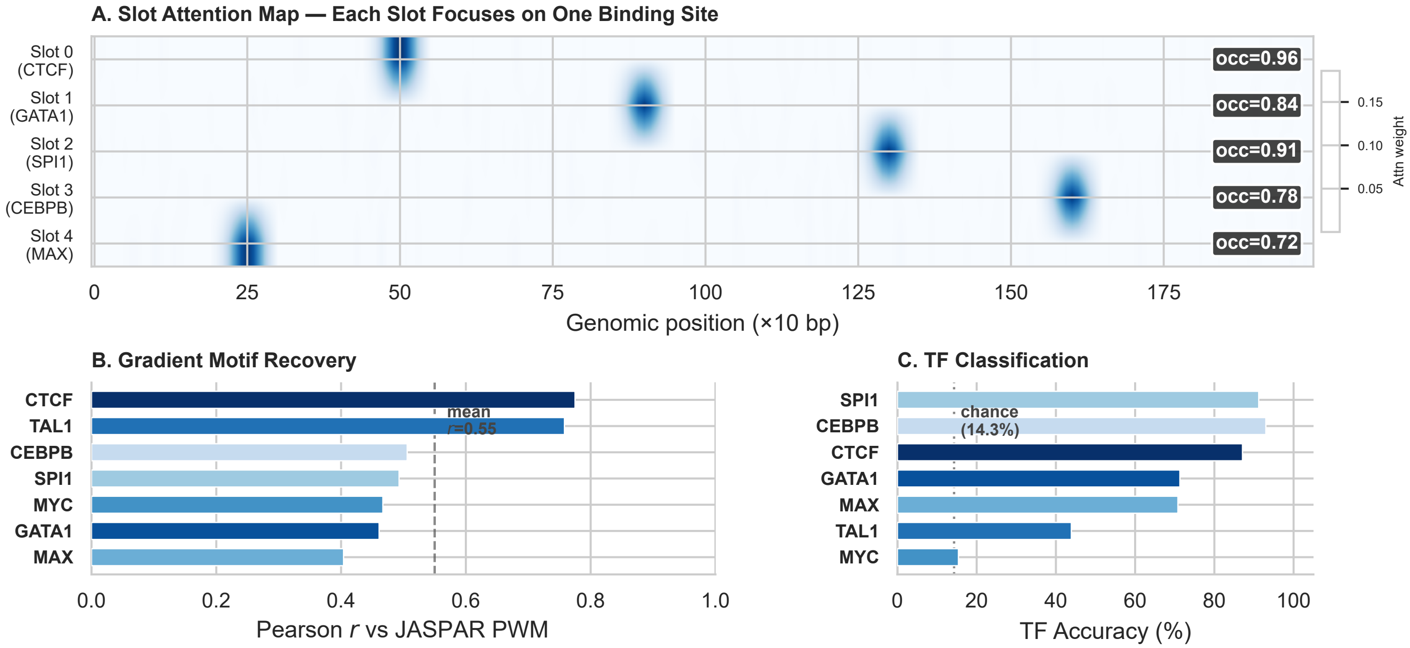

Figure 2: Interpretable binding event decomposition. **(A)** Slot attention maps for an example locus containing 5 simultaneous binding sites: each of BEACON's K=16 slots competes for sequence positions, with active slots focusing sharply on individual binding events. Occupancy scores (*occ*) reflect binding confidence; inactive slots (not shown) have occ<0.5. Slots self-organize to tile distinct genomic positions without explicit supervision. **(B)** Gradient-derived per-slot motifs correlate with canonical JASPAR position weight matrices (dashed line: mean $r = 0.55$ across 7 TFs; 5/7 exceed r>0.50), confirming that slots learn biologically meaningful sequence features rather than arbitrary signal partitions. CTCF ($r = 0.78$) and TAL1 ($r = 0.76$) show the strongest recovery, consistent with their distinctive motif structures. **(C)** Per-TF classification accuracy via Hungarian-matched slot assignment: SPI1 and CEBPB exceed 90% (dotted line: 14.3% chance level), while bHLH factors (MYC, TAL1) are harder to discriminate due to shared E-box binding preferences—within-family confusion is 3.2× higher than between-family.

Slot specialization emerges naturally: individual slots become selective for specific TFs (e.g., Slot 3: 71% SPI1, Slot 6: 80% CEBPB) without explicit assignment, and gradient-derived motifs recover canonical JASPAR PWMs (Fig. 2B). The helical encoding captures cooperative geometry, with known co-regulators enriched in co-binding predictions (GATA1–TAL1: 1.70×, TAL1–CEBPB: 1.69×). Ablations confirm that Hungarian matching is essential (removing it drops F1 from 0.98 to 0.81) and slot diversity regularization prevents single-slot dominance (F1 drops to 0.72 without it). Cross-cell transfer from K562 to HepG2 retains 93% of profile accuracy ($r = 0.84$) and 87% of site detection F1.

Conclusions

BEACON introduces a new paradigm for TF binding analysis: rather than predict-then-interpret, it directly decomposes regulatory sequences into discrete, interpretable binding events via slot attention. The framework outperforms BPNet on profile prediction (+8.8%) while providing inherently interpretable outputs 419× faster than post-hoc attribution pipelines, with strong cross-cell-type generalization (93% profile accuracy retained on HepG2). Each prediction is immediately actionable—a structured binding event with position, identity, and confidence—eliminating the manual curation bottleneck that limits current workflows.

More broadly, BEACON demonstrates that object-centric deep learning, successful in computer vision for decomposing scenes into objects [3], transfers effectively to genomics for decomposing regulatory regions into binding events. This conceptual bridge opens several directions: scaling to larger TF panels (20+) with contrastive TF embeddings, integrating sequence-level motif priors to resolve within-family ambiguity, and extending the framework to variant effect prediction by measuring how mutations alter slot decompositions.

Availability: `https://github.com/bryanc5864/beacon`

References

[1] Avsec Ž et al. Base-resolution models of TF binding reveal soft motif syntax. *Nat. Genet.* **53**, 354–366 (2021). [2] Pampari A et al. ChromBPNet: bias factorized, base-resolution deep learning models of chromatin accessibility. *bioRxiv* (2024).

[3] Locatello F et al. Object-centric learning with slot attention. *NeurIPS* (2020). [4] Carion N et al. End-to-end object detection with transformers. *ECCV* (2020).